

Element C

High Priority Requirements:

1. Safety
 - **Requirement:** No part of the LED lights are in access to the children, besides if we want to make them a control panel
 - **Rationale:** The kids are trusted to be in a safe learning environment, and it is very important that they are unable to get hurt or have that possibility
2. Aesthetic/creates diverse mood
 - **Requirement:** LED lights are able to match moods that we set, and create a nice environment for the students. They are able to have 4-5 setting, and change to all colors of the rainbow
 - **Rationale:** The LED lights purpose is to be able to create setting, whether it calming or supposed to represent a real life setting
3. Durability
 - **Requirement:** LED lights are able to last 4-6 years, and if there are problems left with instructions for fixation
 - **Rationale:** LED lights should be able to help the kids in the class through the year and future kids in the class for multiple years
4. Entertaining and effectiveness
 - **Requirement:** Kids are able to stay engaged for the entirety of activity
 - **Rationale:** We want to create a design that helps the kids stay engaged in class and want to be in class to learn.
5. Adaptability to different needs
 - **Requirement:** Product has the ability to be able to put in its own code, and we are able to add settings
 - **Rationale:** The LED lights can be used for multiple different purposes and goes beyond the surface of what they can do. Teachers can use them depending on varying needs and interests of the students.

Low Priority:

6. Ease of Use
 - **Requirement:** Product is able to be controlled by teachers in the classroom without difficulty
 - **Rationale:** We are not going to be able to be in the classroom while in use, so the teacher has to be the one using it

7. Number of Zones

- **Requirement:** There are at least 3 zones and each zone is able to create the 4-5 settings and all colors of the rainbow (included in the aesthetic)
- **Rationale:** We want the adults to be able to control different parts of the room to achieve different purposes. This makes the LED lights more flexible for use and able to have different learning environments throughout the room.

8. Affordability

- **Requirement:** Under ≈\$150
- **Rationale:** We have to buy a cheap system and code it because these already exist, so we are trying to create our goal out of a cheaper version, and we have a budget (money doesn't grow in trees)

EXTRAS:

- Games
- Kids control panel
- Timers
- Simulations of Real World Environments

Stakeholder Validation

We have discussed with our primary stakeholders to ensure our requirements match our users needs. From our discussions, we have taken away that the most important feature of our design is the zoning and the different colors that are offered. It is also important that we make the lights easy to use for the teachers, as they already have a lot on their hands. Additionally, we received some feedback regarding the safety of the LED lighting system. Specifically, because this classroom is made up of students with all different needs and health requirements, we have to be very weary when creating a design so that it does not adversely impact anyone in the class. This means that we need to do more research on our design so that we can better understand how children's needs may differ within the classroom. The most important thing is that we are helping the class, not harming it, so we need to keep the safety of our students at the forefront of all of our decision making. Overall, from the feedback we received, our project was headed in the right direction, but we needed to dig deeper and make sure that every decision we were making was going to help the students and teachers have a better experience at school. This may entail more research, but it is worth doing the extra work so that our project can be called a success and so that it can help as many people as possible.